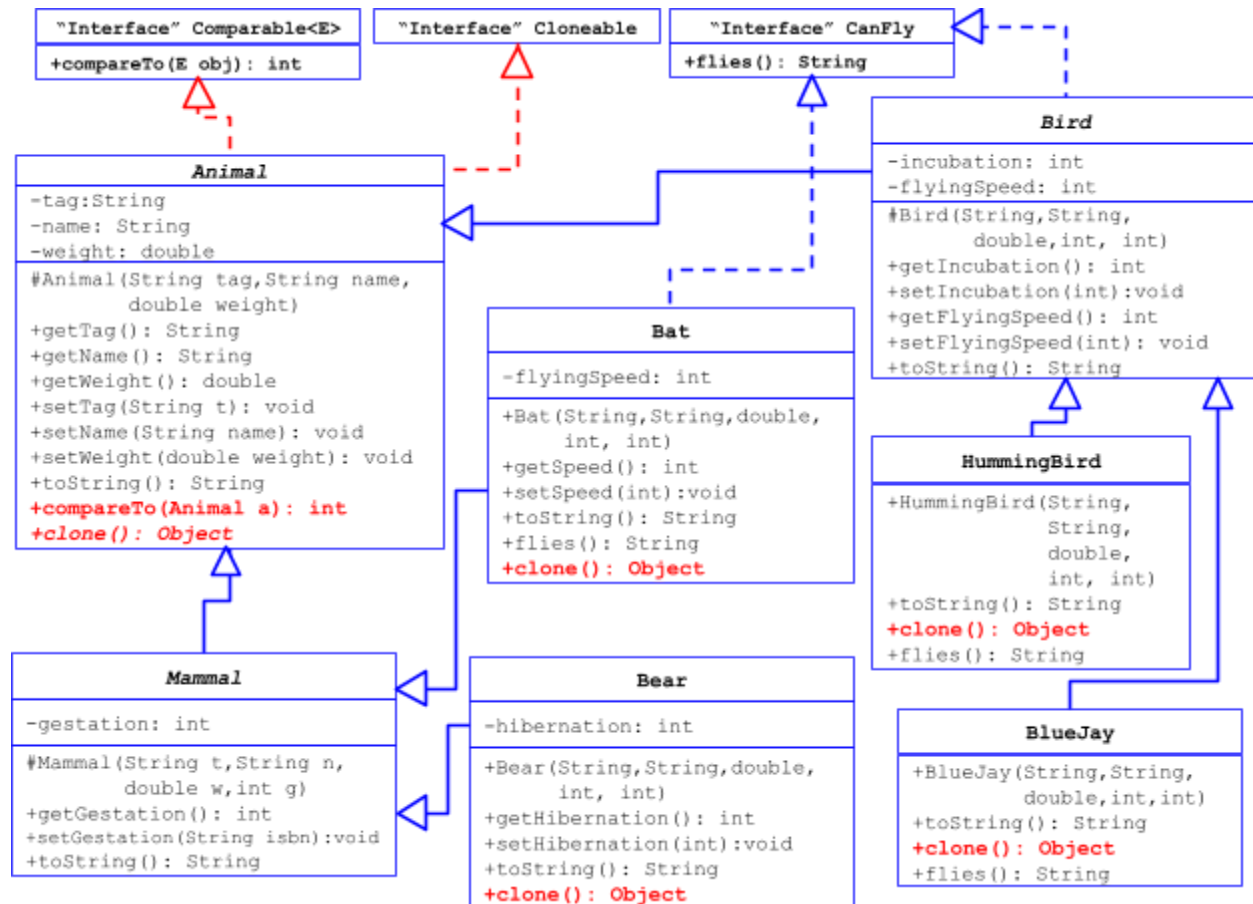


Programming and Data Structures

Test 1 (practice)

The class hierarchy shown in the UML diagram below has been implemented at 80% and the code is provided for you.



You are required to complete the parts described below, which are also highlighted in red on the UML diagram.

Animal Class and Interfaces

The **Animal** class must implement the **Comparable** and **Cloneable** interfaces.

1. Write the definition for the **compareTo** method to order Animal objects based on their weight.
2. Declare the **clone()** method as an abstract method within the **Animal** class.
3. In the concrete classes (**Bat**, **Bear**, **HummingBird**, and **BlueJay**), write the definition of the **clone()** method to return a deep copy of the object being cloned.

Test Class Methods

In the **Test** class, you need to define the following methods:

1. **int readAnimals(Animal[] list, String filename)**: This method reads animal information from the specified text file. It then creates instances of the concrete animal classes and stores them in the **list** array.
 - a. The file contains one animal per line, with attributes separated by a colon (:).
 - b. Use the **split()** method to extract the attributes from each line.
 - c. The attributes are in this order: type, tag, name, weight, gestation or incubation period, and flying speed or hibernation.
 - d. The method should return the number of animals read and stored in the array.
2. **void printFlyingAnimals(Animal[] list, int size)**: This method prints the return value of the **flies()** method for each animal of type **CanFly**.
3. **boolean checkTag(String tag)**: This method validates a user-entered tag.
 - a. A valid tag must be two uppercase letters followed by a hyphen (-) and exactly four digits.
 - b. Use the regular expression "[A-Z]{2}-\\d{4}" to validate the tag.
 - c. If the tag does not match the regular expression, the method should throw an **Exception**, print an error message, and return false.
 - d. If the tag is valid, the method should return true.
 - e. This method must use the catch rule.

Submission and Notes

Submit the following files: **Animal.java**, **Bird.java**, **Mammal.java**, **BlueJay.java**, **HummingBird.java**, **Bear.java**, **Bat.java**, **CanFly.java**, and **Test.java**. Do not modify the **Bird.java** and **Mammal.java** files.

A sample run of the program is provided for you to test your code after adding the required logic.

----- **Sample run 1 (Valid Tag)** -----

File animals.txt read successfully. 12 animals read from the file

List of Animals

Type	Tag	Name	Weight		
Bear	WA-2019	Asian Black	181.44Kg	210	5
Bat	PA-2011	Bulldog	60.00g	25	28
Blue Jay	CA-2002	California scrub jay	87.88g	17	17
Bear	WA-2017	Brown	90.72Kg	200	5
Blue Jay	NY-2005	Pinyon jay	90.00g	19	20
HummingBird	IL-2015	Violetear	7.25g	14	90
Bat	NM-2020	Microbat	5.00g	100	44
HummingBird	WA-2007	Barbthroat	5.00g	18	68
Bear	AL-2011	Polar	226.80Kg	240	4
Blue Jay	NM-2012	Mexican jay	120.00g	18	22
Bear	AL-2012	Polar	226.80Kg	240	0
Bat	FL-2010	HorseShoe	10.00g	91	50
Bear	AL-1119	Polar	226.80Kg	240	6
Bat	PA-9940	Natalidae	60.00g	25	28
Bear	WA-4693	Brown	90.72Kg	200	6
Blue Jay	CA-1807	California scrub jay	87.88g	17	34

List of Flying Animals

Type	Name	Flying Speed (mph)
Bat	Bulldog	28
Blue Jay	California scrub jay	17
Blue Jay	Pinyon jay	20
HummingBird	Violetear	90
Bat	Microbat	44
HummingBird	Barbthroat	68
Blue Jay	Mexican jay	22
Bat	HorseShoe	50
Bat	Natalidae	28
Blue Jay	California scrub jay	34

List of Animals sorted by weight

Type	Tag	Name	Weight		
Bat	NM-2020	Microbat	5.00g	100	44
HummingBird	WA-2007	Barbthroat	5.00g	18	68
HummingBird	IL-2015	Violetear	7.25g	14	90
Bat	FL-2010	HorseShoe	10.00g	91	50
Bat	PA-2011	Bulldog	60.00g	25	28
Bat	PA-9940	Natalidae	60.00g	25	28
Blue Jay	CA-2002	California scrub jay	87.88g	17	17
Blue Jay	CA-1807	California scrub jay	87.88g	17	34
Blue Jay	NY-2005	Pinyon jay	90.00g	19	20
Blue Jay	NM-2012	Mexican jay	120.00g	18	22
Bear	WA-2017	Brown	90.72Kg	200	5
Bear	WA-4693	Brown	90.72Kg	200	6
Bear	WA-2019	Asian Black	181.44Kg	210	5
Bear	AL-2011	Polar	226.80Kg	240	4
Bear	AL-2012	Polar	226.80Kg	240	0
Bear	AL-1119	Polar	226.80Kg	240	6

Enter a tag:

NM-2012

Animal found: Blue Jay	NM-2012	Mexican jay	120.00g	18	22
------------------------	---------	-------------	---------	----	----

----- **Sample run 2 (Invalid Tag)** -----

File animals.txt read successfully. 12 animals read from the file

List of Animals

Type	Tag	Name	Weight		
Bear	WA-2019	Asian Black	181.44Kg	210	5
Bat	PA-2011	Bulldog	60.00g	25	28
Blue Jay	CA-2002	California scrub jay	87.88g	17	17
Bear	WA-2017	Brown	90.72Kg	200	5
Blue Jay	NY-2005	Pinyon jay	90.00g	19	20
HummingBird	IL-2015	Violetear	7.25g	14	90
Bat	NM-2020	Microbat	5.00g	100	44
HummingBird	WA-2007	Barbthroat	5.00g	18	68
Bear	AL-2011	Polar	226.80Kg	240	4
Blue Jay	NM-2012	Mexican jay	120.00g	18	22
Bear	AL-2012	Polar	226.80Kg	240	0
Bat	FL-2010	HorseShoe	10.00g	91	50
Bat	PA-8037	Natalidae	60.00g	25	28
Bear	AL-10891	Polar	226.80Kg	240	6
Bear	AL-9477	Polar	226.80Kg	240	6
Bear	AL-2695	Polar	226.80Kg	240	6

List of Flying Animals

Type	Name	Flying Speed (mph)
Bat	Bulldog	28
Blue Jay	California scrub jay	17
Blue Jay	Pinyon jay	20
HummingBird	Violetear	90
Bat	Microbat	44
HummingBird	Barbthroat	68
Blue Jay	Mexican jay	22
Bat	HorseShoe	50
Bat	Natalidae	28

List of Animals sorted by weight

Type	Tag	Name	Weight		
Bat	NM-2020	Microbat	5.00g	100	44
HummingBird	WA-2007	Barbthroat	5.00g	18	68
HummingBird	IL-2015	Violetear	7.25g	14	90
Bat	FL-2010	HorseShoe	10.00g	91	50
Bat	PA-2011	Bulldog	60.00g	25	28
Bat	PA-8037	Natalidae	60.00g	25	28
Blue Jay	CA-2002	California scrub jay	87.88g	17	17
Blue Jay	NY-2005	Pinyon jay	90.00g	19	20
Blue Jay	NM-2012	Mexican jay	120.00g	18	22
Bear	WA-2017	Brown	90.72Kg	200	5
Bear	WA-2019	Asian Black	181.44Kg	210	5
Bear	AL-2011	Polar	226.80Kg	240	4
Bear	AL-2012	Polar	226.80Kg	240	0
Bear	AL-10891	Polar	226.80Kg	240	6
Bear	AL-9477	Polar	226.80Kg	240	6
Bear	AL-2695	Polar	226.80Kg	240	6

Enter a tag:

AZ-100

Invalid TAG: AZ-100. Must have 2 letters followed by 4 digits